QPT: latest follow-up complete

Verified snapshot: 06 October 2026, 18:55 CEST

Latest: all 12 follow-up cases checked

Full-gate passes: printed Quiroga 0/4; FRAMES 1/2; Armijo control 5/6. Passing calibration cases are not a robust, confirmed comparison. A successful Armijo control is not a Quiroga result.

Completed FRAMES RMSE: 8.427e-07 to 1.111e-06; printed Quiroga RMSE: 0.001341 to 0.001602. Required: <= 1e-6. These remain two calibration channels, not unseen confirmation.

Earlier 60-case pilot (table below): the same corrected n = 2, full-128-row, noiseless rank-one problem, calibration channels 0 and 1, original initial factors and one FRAMES stream. All earlier outcomes remain unchanged.

Earlier 60-case pilot	Cases	Normal	Failed	Final gate
Quiroga printed adaFGD	42	34	8	0
FRAMES	12	12	0	0
Armijo diagnostic control	6	2	4	1

The gate still requires fidelity >= 0.99, full-design RMSE <= 1e-6 and normalized TP error <= 0.001 together. A failed terminal status is never relabeled as a gate pass, even if an earlier or final checkpoint meets the physical thresholds.

What the pilot is resolving

No sign/factor discrepancy was found in the reviewed printed update and TP-gradient equations. Quiroga's printed rule is scale-sensitive: large fixed TP penalties can make updates stiff, while larger step multipliers at low penalties had been underexplored. The expanded grid tests that hypothesis without changing the printed update.

For FRAMES, the true channel is feasible with zero measurement and TP loss. Averaging boundary atoms can shrink the factor norm while trace-normalized fidelity looks excellent. Faster averaging, lower initial penalty and modest radius slack were tested separately; slack-radius runs are domain ablations, not identical-domain comparisons.

Armijo is a recoverability control, not Quiroga or a third primary competitor. Its actual line-search work is not budget-matched; physically accurate checkpoints do not erase later terminal failures.

Armijo-only stopping fix: development status

Merged locally at 0f7857e; not pushed; deployed to a separate pinned Ruche checkout. Full suite: 556 passed, 55 skipped. Review: independently approved; 33 post-merge tests and 18 harness/audit tests also passed. Ruche confirmation: confirmed: four former roundoff-stall controls now converge and pass; 5 of 6 total controls pass.

Repository status: clean, ahead by 4 commits, not pushed; Ruche uses a separate pinned checkout.

This optional stopping change does not alter printed Quiroga or FRAMES, and cannot retroactively change any archived pilot status. Earlier source-fixed validation remains unchanged: Quiroga 0/8 and FRAMES 0/24 passed the full gate.

Separate follow-up: calibration, not new validation

Job 2027604: both tasks COMPLETED, exit 0; last finish 6 October 18:38:10 CEST. Source 0f7857e94f6b.

All 12 follow-up cases are audited; full endpoints and work accounting are on page 3. Best-for-best selection and fresh confirmation remain unfinished.

Per-channel endpoints, not a robustness claim

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FRAMES: every pilot arm, separated by channel

Arm	RMSE c0	TP c0	RMSE c1	TP c1	Gate 0/1
Original	8.711e-06	9.18e-05	0.002567	0.01959	F/F
Faster averaging	4.702e-06	4.927e-05	4.507e-06	5.156e-05	F/F
Lower initial penalty	8.207e-05	0.001729	6.033e-05	0.001251	F/F
Lower + faster	1.383e-05	0.0002852	1.152e-05	0.0002319	F/F
Radius 2.02	3.438e-06	2.57e-05	0.004406	0.0216	F/F
Radius 2.1	6.391e-06	4.298e-05	2.252e-06	1.277e-05	F/F

P/F means terminal full-gate pass/fail, including fidelity and method status. Both channels are calibration cases, not unseen confirmation. One stream per channel does not resolve sampling variability or establish which arm is robust.

Printed Quiroga: descriptive fidelity maxima

Channel	Fixed alpha; multiplier	Fidelity	RMSE	TP	Gate
0	a=5e-06; e=1	0.8276	0.01135	0.07293	F
1	a=5e-06; e=3	0.9846	0.003391	0.01701	F

These are merely the largest completed endpoint fidelities on each channel, not selected configurations or winners. All 42 printed-rule cases, including failures, remain in the complete audit.

Armijo control: all endpoints and retained checkpoint evidence

Channel / fixed alpha	Status	RMSE	TP	Final gate	First physical
c0 a=0.5	complete	3.086e-10	1.117e-12	P	6400
c0 a=0.0005	failed	1.915e-16	5.212e-16	F	1100
c0 a=5e-05	failed	1.516e-16	7.36e-16	F	1100
c1 a=0.5	complete	4.362e-06	1.222e-07	F	None
c1 a=0.0005	failed	1.887e-16	4.567e-16	F	1100
c1 a=5e-05	failed	1.759e-16	8.411e-16	F	1100

First physical = first archived iteration meeting all physical thresholds, retrospectively; it is not an early-stopped result. Late Armijo failures remain failed scientific outcomes, not infrastructure failures or Quiroga successes.

All previous calibration/validation evidence is preserved. No held-out data or validation retuning was used. The original initialization remains; equivalence to the paper's Gell-Mann-basis initialization is still open, so this is not an exact reproduction of every published choice. Revised configurations require fresh confirmation.

Follow-up settings (page 3): Quiroga alpha is fixed, with step multiplier 1. FRAMES alpha is the initial 5e-5 and grows as (k+1)^0.25; its averaging step is 4/(k+4), radius 2 and batch size 32. Armijo uses full-gradient stopping atol = 1e-12 and rtol = 1e-10, separately from the physical gate.

Full 60-case evidence: audit_complete.json

Primary paper: arXiv:2312.01311v1

Longer-horizon follow-up: all 12 cases

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Separate calibration evidence, with the new Armijo-only stop

All 12 follow-up cases and both inventories are verified against the new pinned source and the original pilot's data and initial factors. Printed Quiroga and FRAMES update equations are unchanged. Only the Armijo control opts into the reviewed stationarity stop; recovery still requires a separate physical-gate check.

Quiroga printed adaFGD

Channel / alpha	End step	Status	Fidelity	RMSE	TP	Gate	First step
c0 a=0.0005	100000	complete	0.9965	0.001602	0.003006	F	None
c0 a=5e-06	100000	complete	0.9967	0.001546	0.007114	F	None
c1 a=0.0005	100000	complete	0.9969	0.001427	0.002502	F	None
c1 a=5e-06	100000	complete	0.9973	0.001341	0.005953	F	None

FRAMES

Channel / alpha	End step	Status	Fidelity	RMSE	TP	Gate	First step
c0 a=5e-05	400000	complete	1	1.111e-06	9.523e-06	F	None
c1 a=5e-05	400000	complete	1	8.427e-07	7.953e-06	P	308000

Armijo diagnostic control

Channel / alpha	End step	Status	Fidelity	RMSE	TP	Gate	First step
c0 a=0.5	7633	converged	1	5.634e-08	2.166e-10	P	6400
c0 a=0.0005	1853	converged	1	1.141e-10	2.071e-10	P	1100
c0 a=5e-05	1794	converged	1	8.968e-11	3.501e-10	P	1100
c1 a=0.5	10000	complete	1	4.362e-06	1.222e-07	F	None
c1 a=0.0005	2026	converged	1	1.239e-10	2.224e-10	P	1100
c1 a=5e-05	1992	converged	1	8.064e-11	3.02e-10	P	1100

End step and first step are native optimizer iterations, not equal units of work across methods. Printed Quiroga uses up to 100,000 full-data updates; FRAMES uses 400,000 batch-32 updates (12.8 million nominal rows each when complete). Armijo uses at most 10,000 iterations with separately counted line-search and stopping-check work. Primary nominal row counts exclude Quiroga spectral-numerator work and checkpoint metrics; they are not CPU-time equivalents.

First step is the first archived checkpoint meeting all physical thresholds; it is retrospective and does not replace terminal status. P/F is the terminal full gate, including status, fidelity, RMSE and TP. Old failed Armijo results remain failed.

More iterations do not promise recovery. Two reused calibration channels and one FRAMES stream do not establish a winner or robustness; no held-out data were used. Next: reconcile unresolved paper initialization and step conventions, and predeclare a Gell-Mann-basis initialization control before much longer Quiroga runs. No error in the published method has been established.

Full follow-up evidence: audit_complete.json